

MCQs (K-Nearest Neighbors, Logistic Regression)

1. Which distance metric is used by default in `KNeighborsClassifier()`?
 - a) Euclidean
 - b) Manhattan
 - c) Hamming
 - d) Cosine

Ans: a)

Explanation: The default distance metric used in `KNeighborsClassifier` is `minkowski`, which when set with `p=2` is equivalent to the standard Euclidean distance. Euclidean distance metric is used by default in `KNeighborsClassifier()` as per this question.

2. KNN is considered a **non-parametric algorithm** because:
 - a) It does not make any assumptions about the data distribution
 - b) It requires tuning parameters
 - c) It uses probability distributions
 - d) It builds a decision boundary before training

Ans: a) It does not make any assumptions about the data distribution

Explanation: KNN is considered a **non-parametric algorithm** because it doesn't make any assumptions about the data distribution.

3. Which of the following activation functions is used in Logistic Regression?
 - a) ReLU
 - b) Sigmoid
 - c) Softmax
 - d) Tanh

Ans: b)

Explanation: Sigmoid activation function is used in logistic regression.

4. In Logistic Regression, **higher values of the regularization parameter (C)** will:
 - a) Reduce overfitting
 - b) Increase overfitting
 - c) Have no effect
 - d) Make the model faster

Ans: b)

Explanation: In Logistic Regression, higher values of the regularization parameter (C) will lead to less regularization; meaning the model will be allowed to fit the training data more closely, potentially increasing the risk of overfitting.

5. The regularization technique used in scikit-learn's **LogisticRegression()** by default is:
 - a) L1 (Lasso)
 - b) L2 (Ridge)
 - c) Elastic Net
 - d) No regularization

Ans: L2 (Ridge)

Explanation: In scikit-learn, the default regularization technique used in the `LogisticRegression()` function is L2 regularization.

Week 8 Assignment

1. What is the primary principle behind Occam's Razor in model selection?

- a) Always prefer the most complex model
- b) Simpler models should be preferred if they provide the same generalization error
- c) Select the model with the highest number of parameters
- d) Choose the model that fits the training data perfectly

Answer: b) Simpler models should be preferred if they provide the same generalization error

Explanation: Occam's Razor suggests that, given two models with similar generalization errors, the simpler one should be preferred as it is expected to generalize better.

2. Why is Occam's Razor not always applicable to ensemble learning models?

- a) Ensemble models are always inaccurate
- b) Simpler models always outperform complex models
- c) Empirical studies show that increasing model complexity in ensembles can reduce generalization error
- d) Occam's Razor does not apply to machine learning

Answer: c) Empirical studies show that increasing model complexity in ensembles can reduce generalization error

Explanation: Although Occam's Razor suggests simpler models generalize better, studies have shown that ensemble methods like boosting can improve performance despite increased complexity.

3. What is the key difference between Occam's First Razor and Second Razor in machine learning?

- a) The first razor prioritizes simpler models when generalization error is equal, while the second razor assumes simpler models generalize better even with equal training error
- b) The first razor prefers complex models, and the second razor prefers simple models
- c) Both razors suggest the same thing
- d) The first razor applies only to deep learning

Answer: a) The first razor prioritizes simpler models when generalization error is equal, while the second razor assumes simpler models generalize better even with equal training error

Explanation: The first razor is widely accepted as a good heuristic, whereas the second razor is often not supported by empirical evidence.

4. How is pessimistic error estimated in decision trees?

- a) By using the training error directly
- b) By incorporating a complexity penalty term
- c) By maximizing the generalization error
- d) By using the test error only

Answer: b) By incorporating a complexity penalty term

Explanation: The pessimistic error estimate adjusts the training error by adding a penalty for model complexity to prevent overfitting.

5. Why is model complexity considered while evaluating a model?

- a) More complex models always have lower error
- b) Complex models are preferred for interpretation
- c) Complex models are prone to accidental fitting, leading to poor generalization
- d) Simpler models cannot capture patterns

Answer: c) Complex models are prone to accidental fitting, leading to poor generalization

Explanation: Overly complex models may fit the training data too well, capturing noise rather than the actual pattern, resulting in poor performance on new data.

6. What type of learning is K-Nearest Neighbor (K-NN)?

- a) Instance-based learning
- b) Parametric learning
- c) Deep learning
- d) Reinforcement learning

Answer: a) Instance-based learning

Explanation: K-NN is a **lazy learning algorithm** where the model does not learn a decision boundary during training but instead stores training data and performs classification at prediction time.

7. What distance metric is most commonly used in K-NN classification?

- a) Manhattan distance
- b) Euclidean distance
- c) Hamming distance
- d) Cosine similarity

Answer: b) Euclidean distance

Explanation: Euclidean distance is widely used in K-NN to measure the closeness between a test instance and training examples.

8. Why is K typically chosen as an odd number in K-NN?

- a) To ensure deterministic decision-making and avoid ties
- b) Because even numbers increase complexity
- c) Because K-NN cannot work with even numbers
- d) To maximize training accuracy

Answer: a) To ensure deterministic decision-making and avoid ties

Explanation: Choosing an odd K prevents ties in majority voting, ensuring a definite class label assignment.

9. How does weighted K-NN improve over basic K-NN?

- a) By assigning more weight to distant neighbors
- b) By assigning more weight to closer neighbors
- c) By ignoring the training data
- d) By using deep learning for classification

Answer: b) By assigning more weight to closer neighbors

Explanation: Weighted K-NN gives more influence to closer neighbors, improving classification accuracy in cases where distance matters.

10. What is a key disadvantage of K-NN?

- a) It is computationally expensive as it requires calculating distances for every prediction
- b) It does not require training data
- c) It always overfits the data
- d) It cannot handle large datasets

Answer: a) It is computationally expensive as it requires calculating distances for every prediction

Explanation: Since K-NN stores all training data and computes distances at prediction time, it can be slow, especially for large datasets.
